Q 9 Write a program to create a class book having members id, name and price. perform accept and display operation for 2 books using pointer to object.

→

```
#include <iostream.h>
#include <conio.h>
#include <stdio.h>
class book
{
    public:
    int id;
    char name[50];
    int price;

    void accept()
    {
     cout<<" Enter the ~~title~~ ID of book: ";
     cin>>id;
     cout<<" Enter the name of book:";
     gets(name);
     cout<<" Enter the price of book:";
     cin>> price;
    }

    void display()
    {
     cout<<" ID is : "<< id<< endl;
     cout<<" Name is :  "<<name <<endl;
     cout<<" Price is :  "<< price<< endl;
    }
};
```

```
void main()
{
    book b1, *b2, b3, *b4;
    b2 = &b1;
    ~~b3 = &b4;~~
    b4 = &b3;
    b2 -> accept();
    b2 -> display();
    b4 -> accept();
    b4 -> display();
    getch();
}
```

Q.10. Write a program to overload binary operator to concat 2 strings.

→

```
#include <iostream.h>
#include <conio.h>
#include<string.h>
#include <stdio.h>

class name
{
 char str1[10];
 public:

 void getdata()
 {
  cout<<"\n Enter a strings";
  ~~cin~~ gets(str1);
 }
```

```
void operator +(name n)
{
    cout<< strcat (str1, n.str1);
}
};
void main()
{
 name n1, n2;
 clrscr();
 n1. getdata();
 n2. getdata();
 n1 + n2;
 getch();
}
```

Q. 11 Write a program to overload unary operator to negate values.

→

```
#include <iostream.h>
#include <conio.h>
#include <stdio.h>
#include <string.h>
class Number
{
    int x, y;
    public:
    Number (int a, int b)
    {
        a = x;
        b = y;
    }
```

```
void display()
{
  cout<<" value of x: "<<x;
  cout<<" value of y: "<<y;
}
void operator -()
{
  x = -x;
  y = -y;
}
};
void main()
{
  Number N1 (5,6);
  clrscr();
  N1.display();
  -N1;
  cout<<" \n After negation:";
  N1.display();
  getch();
}
```

Q. 12 Write a program to implement friend function.

→

```
#include <iostream.h>
#include <conio.h>
#include <stdio.h>
class second;
class first
{
  int x;
  public:
```

```
void getx()
{
    cout<<"\n Enter the value of x:";
    cin>>x;
}
friend void max (first ,second);
};

class second
{
    int y;
    public:
    void gety()
    {
    cout<<"\n Enter the value of y:";
    cin>>y;
    }
friend void max (first ,second);
};
void max (first a , second b)
{
    if (a.x >b.y)
    {
    cout<<" Greater value is :"<<a.x;
    }
    else
    {
    Cout<<" Greater value is :"<<b.y;
    }
}
```

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```
void main()
{
  First a;
  second b;
  clrscr();
  a.get x();
  b.get y();
  max (a,b);
   getch();
}
```

Q. 13 Write a program to implement following inheritance.

→

```
#include <iostream.h>
#include <conio.h>
#include <stdio.h>
 class student
{
    public:
    int rno;
    char name [50];
};
class test : public student
{
    public:
    int marks1, mark2;
};
class sports
{
    public:
    int score;
};
```

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```
class result : public sports, public test
{
  public:
  int total;

  void accept()
  {
   cout<<"\n Student Information";
   cout<<"\n Enter roll no : ";
   cin>>rno;
   cout<<"\n Enter the ~~m~~ name:";
   gets(name);
   cout<<"\n Enter the marks1 :";
   cin>> marks1;
   cout<<"\n Enter the marks2 :";
   cin>> marks2;
   cout<<"\n Enter the score :";
   cin>> score;
  }

  void display()
  {

   cout<<" Roll No : "<<rno<<endl;
   cout<<" Name is : "<<name<<endl;
   cout<<" Score is : "<<score<<endl;
   ~~cout<~~
   total = marks1 + marks2 + score;
   Cout<<" total is : "<<total<<endl;
  }
};
```

```
void main()
{
  clrscr();
  result s1;
  s1.accept();
  s1.display();
  getch();
}
```

Q 14 Write a program to implement following inheritance.

→

```
#include <iostream.h>
#include <stdio.h>
#include <conio.h>
class staff
{
    public:
    int code;
};
class teacher : public staff
{
    public:
    char subject [50];

    void accept()
    {
     cout<<" \n Enter the code:";
     cin>> code;
     cout<<" \n Enter the name of subject:";
     gets(subject);
    }
```

```
void display()
{
  cout<<"\n code is: "<<code <<endl;
  cout<<"\n Subject is: "<<subject<<endl;
 }
};
 class officer : public staff
{
   public:
   char grade [50];

   void accept()
   {
    cout<<"\n Enter the grade:";
    gets(grade);
   }
   void display()
{
   cout<<"\n Grade is: "<<grade << endl;
 }
};
 void main()
{
  clrscr();
  teacher t1;
  officer o1;
  t1.accept();
  o1.accept();
  t1.display();
  o1.display();
  getch();
}
```

Q15 Write a program to implement following inheritance

→

```
#include <iostream.h>
#include <conio.h>
#include <stdio.h>
class collegestudent
{
    public:
    int st_id;
    int clg_code;
};
class test: virtual public collegestudent
{
    public:
    int per;
};
class sports : virtual public collegestudent
{
    public:
    char grade;
};
class result: public test, public sports
{
    public:
    void accept()
    {
        cout<<" \n Enter the student id :";
        cin>> st_id;
        cout<<" \n Enter the class code:";
        cin>> clg_code;
```

```
cout<<"\nEnter the percentage:";
cin>>per;
cout<<"\n Enter the grade:";
cin>>grade;
}
void display()
{
 cout<<"\n Student ID:"<<st_id<<endl;
 cout<<"\n class code :"<<clg_code<<endl;
 cout<<"\n Percentage : "<<per<<endl;
 cout<<"\n Grade : "<<grade<<endl;
 }
};
void main()
{
  clrscr();
  result rt;
  rt.accept();
  rt.display();
   getch();
}
```

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